Elias Hossain

Open to Internship Opportunities in Applied Science, Research, and Machine Learning Engineering

Flexible start date; Ph.D. coursework completed

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Research Profile

Ph.D. candidate (expected May 2028) with **3 years of industry experience in machine learning**, including deploying speech recognition in a production e-governance system. Research focuses on efficient, training-free approaches to post-training and agent reliability, including steering frozen policies without retraining, sub-millisecond runtime safeguards for tool-using agents, and inference-time calibration. **Publications in NeurIPS 2026, EACL 2027 (Main Track), and Findings of AACL-IJCNLP 2026.**

Education

University of Central Florida, Orlando, FL — Aug 2025 – May 2028 (Expected)
Ph.D. in Industrial Engineering (CS concentration), GPA 3.87/4.00
Advisor: Dr. Niloofar Yousefi | *Qualifying Exam: Passed*

Mississippi State University, Mississippi State, MS — Jan 2024 – Jul 2025
M.S. in Computer Science (Research), GPA 3.88/4.00
Thesis: "TopNet R1: A Multi-Stage AI Framework for Topic Discovery in Scientific Abstracts" | *Advisor: Dr. Andy Perkins*

Research and Technical Experience

Graduate Research Assistant — Aug 2025 – Present
Complex Adaptive Systems Laboratory, University of Central Florida — *Orlando, FL*

- Built multi-agent preference-data verification, reducing Qwen2-7B summarization refusals from **14.8% to 0.5%** without reducing win rate.
- Lead research on post-training policy steering for offline RL, inference-time uncertainty calibration, runtime safety for tool-using LLM agents, and multi-agent security; six papers submitted or accepted (see Publications).
- Released 5 public ML benchmarks (**74,590+ records**) with code.
- Trained and evaluated Qwen, Llama, and Mistral models from 1.5B to 8B parameters on UCF's Newton HPC cluster and a 2× NVIDIA TITAN RTX workstation.

Graduate Research Assistant — May 2025 – Jul 2025
Mississippi State University — *Mississippi State, MS*

- Developed contrastive-learning and uncertainty-estimation methods for detecting rare malicious events in imbalanced security logs.
- Implemented FAISS retrieval and RAGAS evaluation for BarkPlug V.2, a retrieval-augmented assistant deployed to the campus community.

Software Engineer — Dec 2022 – Nov 2023
REVE Systems — *Dhaka, Bangladesh*

- Reduced speech-recognition word error rate by **20%** and inference time by **15%** using DeepSpeech and Wav2Vec 2.0.
- Deployed the models in a production e-governance voice interface.

AI/ML Researcher — Oct 2020 – Nov 2022
Time Research & Innovation (Timerni) — *Remote / UK*

- Developed deep-learning models for COVID-19 screening from chest X-ray and CT images.
- Contributed ML to a telemedicine and health-advice platform (web, iOS, Android) connecting patients, doctors, pharmacies, and hospitals; co-authored the system paper at *ICICT 2021* (Springer LNNS).

Selected Publications and Preprints

[1] **When Policies Cannot Be Retrained: A Unified Closed-Form View of Post-Training Steering in Offline Reinforcement Learning**. **E. Hossain**, *NeurIPS 2026, Main Conference Track (Accepted)*. arXiv Project Code
Adapts frozen offline-RL actors at deployment with no retraining; degrades gracefully under poor priors across 4 D4RL tasks.

[2] **NEXUS: Structured Runtime Safety for Tool-Using LLM Agents**. **E. Hossain**, *Findings of AACL-IJCNLP 2026 (Accepted)*. arXiv Project Code
0.205 ms median latency (under 0.1% overhead), with 0% indirect prompt-injection success at 99% benign allow.

[3] **ChannelGuard: Safe Models Do Not Compose into Safe Multi-Agent Systems**. **E. Hossain**, *ACL ARR (Under Review)*. arXiv Project
Training-free gates block tool poisoning 30/30 across 3 backends; halve prompt-injection success, GSM8K accuracy unchanged.

[4] **UAT-LITE: Inference-Time Uncertainty-Aware Attention for Pretrained Transformers**. **E. Hossain**, *EACL 2027 Main Conference Track (Accepted)*. arXiv Project Code
Inference-time only, pretrained weights unchanged: about 20% lower calibration error vs. BERT-base, accuracy preserved.

[5] **Right Knowledge, Wrong Answer: Test-Time Steering for Temporal Fact Conflicts in Open-Weight Language Models**. **E. Hossain**, *AAAI 2027 (Under Review)*. arXiv Project Code
Verified 8,746-transition Wikidata benchmark; activation patching flips 72–85% of cases across four open-weight models.

[6] **Agreement Overstates Evidence: Error Dependence in LLM Judge Consensus**. **E. Hossain**, *ICLR 2027 (Under Review)*. arXiv Project Code
Ten LLM judges carry about as much information as 3.5 independent ones; shared errors flip up to 28% of significance calls.

Earlier work (2021–2023): 4 papers on applied ML and NLP, including first-author papers at *IEEE ICCIT 2022* (topic modeling and sentiment classification for eco-friendly travel) and *Intelligent Automation & Soft Computing* 2023 (news modeling and retrieval). Full list on Google Scholar.

Research in Progress

- **Adaptive Compute Allocation for Vision-Language Models** *In preparation for CVPR 2027*
  Test-time compute in Qwen2.5-VL-3B: visual resolution, reasoning budgets, and adaptive routing, with cross-fitted oracle analysis of when dynamic allocation helps.
- **LLM Agents for Automated Model Deployment Optimization** *In preparation for ICML 2027*
  An LLM optimization agent for diffusion-model deployment, benchmarked against TPE, hierarchical Bayesian optimization, and random search, including hybrid LLM–Bayesian methods and held-out generalization.

Selected Technical Projects

- **MiniHealthLM** Code: Built a ∼600M-parameter Qwen-style Transformer from scratch in PyTorch (GQA, RoPE, RMSNorm) with medical-text preprocessing, PII filtering, custom BPE tokenization, 1024-token context, and checkpointing.
- **MLDriftGuard** Code: Installable Python package for numerical and categorical drift detection (KS and chi-squared tests) with visualizations and mitigation suggestions. *SciPy, pandas*

Technical Skills

RL & Post-training	RLHF, PPO, GRPO, DPO, SFT, preference learning, offline RL
LLM Systems & Scaling	PyTorch, Hugging Face Transformers, vLLM, LoRA/QLoRA, multi-GPU training
Retrieval & Evaluation	RAG, FAISS, RAGAS, uncertainty calibration, model evaluation
Tools	Python, SQL, R, Git, Docker, Azure, GCP

Service

- Reviewer, NeurIPS 2026 Workshop on Trustworthy AI for Good
- Reviewer, ICML 2026 Workshop on Generative and Agentic AI for Biology